Final Research Report

PDF edition of preserved repository document | docs/final_research_report.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Final frozen coding-study report

Verified coverage: 1920 learning / 400 lockbox / 400 independent audits / 16 fresh latency cases. Original1634-row checkpoint and original1122CPU-row prefix remain byte-identical. Recovery archive is preserved. Neural PID1620 exited; supervisor completed every checked subprocess.

Observed outcomes

Condition	Split	Successes /50	95% Wilson interval
model_only	test	8/50	0.083–0.285
model_only	ood_test	12/50	0.143–0.374
structured	test	50/50	0.929–1.000
structured	ood_test	22/50	0.312–0.577
mindscape_b	test	45/50	0.786–0.957
mindscape_b	ood_test	24/50	0.348–0.615
mindscape_c	test	49/50	0.895–0.996
mindscape_c	ood_test	18/50	0.241–0.499

Task-paired statistics and complete learning cells are in experiments/coding_completion_v2/final_summary.json. Intervals are descriptive and unadjusted for multiple comparisons. Successful replay verifies evidence, not successful repair. Five zero-tool trajectories have undefined tool-result agreement.

All dedicated latency observations

Sequential, cache-disabled MPSfloat16; no model reload per case. Stage intervals can overlap. Four task/condition pairs per condition; no repeated key counted twice.

Condition	Task	Split	Success	Wall seconds
model_only	fd23fa32e4f9	test	False	3.030
model_only	e5545f5d719e	ood_test	False	2.800

Condition	Task	Split	Success	Wall seconds
model_only	34d9c496b646	test	False	2.257
model_only	f7f984138799	ood_test	False	1.674
structured	fd23fa32e4f9	test	True	2.470
structured	e5545f5d719e	ood_test	False	3.142
structured	34d9c496b646	test	True	2.641
structured	f7f984138799	ood_test	False	3.241
mindscape_b	fd23fa32e4f9	test	True	33.891
mindscape_b	e5545f5d719e	ood_test	False	45.430
mindscape_b	34d9c496b646	test	True	37.958
mindscape_b	f7f984138799	ood_test	False	33.686
mindscape_c	fd23fa32e4f9	test	True	26.679
mindscape_c	e5545f5d719e	ood_test	False	32.615
mindscape_c	34d9c496b646	test	True	31.098
mindscape_c	f7f984138799	ood_test	True	32.388

Validation and reproducibility

Fresh original and clean-restored suites each passed 130tests plus6subtests, zero failures/errors/skips. Scoped lint, compilation, wheel build, fairness/training checks and Nodeplayback verification passed. Snapshot results/final/coding_research_v2 and restored file hashes independently verified:6242 files each. Full commands/outcomes, all16 stage measurements, scientific pins and SHA256checksums are in experiments/coding_completion_v2/release_manifest.json.

Limitations

- Mixed hardware/precision and unmatched supervision/interaction compute prevent architecture-only causal attribution

- Synthetic bounded repair tasks; unknown foundation pretraining exposure

- 16latencycases are descriptive, not broad performance estimates

- No measured zero-budget/control component memory/dream/recovery intervention

- Foundation weights are pinned external downloads, excluded from package

- No configured type-check acceptance task; no visual browser QA claimed

- No GPT/Gemini superiority; negative and unsuccessful cases retained

Completion refers to frozen experimental coverage. Broad capability and GPT/Gemini superiority are unsupported; data-efficiency and unmeasured intervention claims remain inconclusive. Package publication/tag verification is recorded separately after export.